Advanced Programming Practice · 21CSC203P

Unit 5 · Session 2 · SLO 1 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 2 : SLO â€“ 1 | Constructing DFA and NFA for Basic Languages

I. Questions & Answers

1. DFA for strings over {0,1} ending with '0' â€” states {q0, q1}; start = q0, accepting = {q1}. Transitions: q0 --0--> q1, q0 --1--> q0, q1 --0--> q1, q1 --1--> q0. The machine is in q1 exactly when the last symbol read was 0.

2. NFA for strings starting with '1' â€” states {q0, q1}; start = q0, accepting = {q1}. Transitions: q0 --1--> q1, and q1 loops on 0 and 1. (From q0, input 0 has no transition â€” the string is rejected.)

3. Can an NFA have Îµ-transitions? â€” Yes. An Îµ-NFA may change state without consuming any input symbol; Îµ-transitions are the key tool in Thompson's construction, and every Îµ-NFA can be converted to an equivalent DFA.